# Educational Game Learning Quality Evaluation Tool: A Systematic Literature Review

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***Abstract* - This Systematic Literature Review (SLR) aims to examine evaluation tools for measuring the quality of Digital Educational Games (DEG) and identify the most popular evaluation tools and significant evaluation factors. This research uses the PRISMA method to search for journals at IEEE, Springer, and Elsevier publishers from November 8, 2023, to December 19, 2023, with specific keywords. Of the 3,808 journals found, 139 journals published from 2020 to 2023 were selected and analyzed using Excel to collect data such as year of publication, title, author, publisher, tool name, tool type, and assessment factors. Of the 139 articles extracted, 35 journals used the DEG evaluation tool. The classification results based on the similarity of evaluation tools identified 12 DEG evaluation tools, including MEEGA+, EgameFlow, TAM, ADDIE, GUESS, HEP, TAUT, EDUGXQ, SPIKES, UEQ, DIJS, and ECS. MEEGA+ was found to be the most frequently used evaluation tool, with 12 references or 23% of the total journals reviewed. Further evaluation using the Analytical Hierarchy Process (AHP) identified control and feedback as evaluation factors that needed further development. This research emphasises the need to develop a more comprehensive and valid DEG evaluation tool to improve the measurement of the quality of digital educational games. In conclusion, these findings provide a solid basis for decision-making regarding DEGs and encourage further research to address gaps in existing evaluation tools.**

***Keywords - Educational games, Evaluation tools, Game design, Serious games*.**

## I. INTRODUCTION

DEG has emerged as a new paradigm in education, implemented based on the concept that children can gain a better understanding through learning that involves problem-solving in a fun atmosphere [1].

Technological developments positively impact children's education using DEG in various disciplines [2]. This DEG is designed to provide learning in certain areas, develop concepts, accelerate development, and help students hone specific skills and approaches during the playing process. Over time, DEG has become an innovative strategy for increasing the effectiveness of learning at higher levels of education [3], with positive impacts seen on academic achievement [4], as well as cognitive development, such as attention and memory skills [5].

Assessments and study reports show that the main aim of DEG development is to support the education sector [6]. In addition, games can increase students' creativity and learning motivation [7] and stimulate involvement, interest, and learning effectiveness [8]. Although DEGs are experiencing rapid development, most have not confirmed their effectiveness as a tool to achieve these goals. Several previous studies were hampered by inadequate research design so the results could not provide valid evidence regarding the effectiveness of using games in responding to various problems [9].

Given the complexity of digital gaming environments and designing games with non-entertainment (educationally) objectives, an urgent need arises to select and compare appropriate evaluation methods [10]. Poor understanding of this important issue is the main motivation behind this research. Identifying more consistent and uniform patterns in various dimensions of the systematic evaluation of DEGs is the basis for achieving more valid evaluation results, which can later become the basis for consideration in decision-making regarding DEGs. Without a valid educational methodological approach, developments in educational design regarding learning environments to achieve complex skills in various fields of knowledge will be limited [11].

Therefore, this systematic review aims to investigate the various evaluation tools used in the context of DEGs to answer the question regarding which evaluation tools can be relied on to evaluate DEGs more validly regarding their development.

## II. METHOD

This SLR was carried out by detailing previous evidence and similar research, aiming to provide detailed answers to the research questions, namely:

**RQ1**. What is the distribution of journals based on publishers used in discussing DEG evaluation tools?
**RQ2**. How many evaluation tools were obtained from the processed journals?
**RQ3**. What evaluation models are popularly used to evaluate DEGs?
**RQ4**. Are there similarities in the evaluation tools used in the journals?
**RQ5**. Are there similarities in evaluation factors between existing evaluation tools?
**RQ6**. Is there an evaluation tool that has evaluation factors that are used throughout the evaluation tool?

This systematic review involves searching for relevant research by searching for related keywords in the title, abstract, and research keywords in several scientific databases, including IEEE, Springer, and Elsevier.

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The scientific journal search process was carried out within the specified period, starting from November 8 2023, to December 19, 2023. The selection of this date was adjusted to the relevant period to obtain the latest information regarding DEG evaluation. The keywords used in this search involve selecting key terms that are appropriate to the research focus, namely:

1. ("Game Design" OR "Educational Games")
2. ("Digital game-based learning" OR "Digital games" OR "Serious games")
3. ("Evaluation" OR "Educational Games")

The article search steps in this study refer to the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guidelines [12]. The data collection process involves the use of a uniform checklist to extract information such as year of publication, author, title, publisher, field of publication, keywords, research object, research problem, research objectives, research methods, research results, contributions, assessment indicators, conclusions and findings or notes.

This approach is directed at caution and consistency so that the data taken is reliable and relevant to the research objectives. This method ensures that the articles used in this research have passed strict selection, comply with established criteria, and can significantly contribute to understanding DEG evaluation tools.

The criteria for including journals in the analysis include several things. First, the journal publication year must be between 2020 and 2023 to ensure the relevance and up-to-dateness of the data. Second, journal research topics must focus on evaluating the quality of Digital Educational Games (DEG) or be directly related to DEG evaluation tools. Third, journals must have a clear and detailed methodology for using evaluation tools in the context of DEGs. Fourth, the journal must be fully accessible to allow in-depth analysis.

On the other hand, the criteria for excluding journals from the analysis are also clearly defined. First, journals published before 2020 will be excluded. Second, journals that do not focus on DEGs or do not specifically discuss evaluation tools will be excluded. Third, journals that do not provide sufficient methodological explanations regarding evaluating tools will also be excluded. Fourth, journals that are duplicates or very similar to other articles already included in the analysis will be deleted. Finally, journals that were not fully accessible, thereby limiting in-depth analysis, were excluded.

With these criteria, journal selection was carried out strictly to ensure that only the most relevant and quality journals were included in the analysis, supporting the validity and reliability of the research results. Fig. 1 shows the percentage of publisher journals obtained.

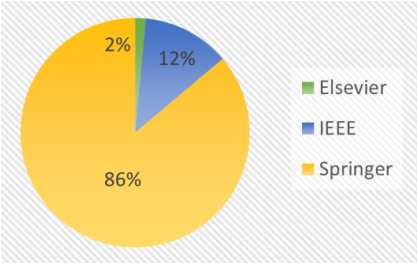



Fig. 1. Percentage of Publisher Journals

In this research, of the 3,808 articles identified from various publishing sources, namely Elsevier (57 articles or 2%), IEEE (471 articles or 12%), and Springer (3,280 articles or 86%), a strict selection process was carried out to ensure relevance. and quality. A total of 3,669 articles that were not relevant, namely outside the publication year 2020-2023, were removed using the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) procedure.

After this process, an initial review was carried out based on the article publication year, title, publisher, and abstract, which resulted in 139 selected articles. Further evaluation of the content of these 139 articles found that 104 articles did not explain the tools used in the research and were therefore excluded from further analysis. Discussions and summaries of the remaining 35 articles were carried out based on similarities in the Digital Educational Games (DEG) evaluation tool.

This sample size of 139 articles can represent the overall research landscape, given the strict and transparent selection procedures, distribution of articles from leading publishers, and focus on the most recent articles from 2020-2023. Thus, this sample can provide a fairly accurate and in-depth picture of DEG evaluation tools in current research.

An overview of the literature search process and study selection can be seen in Fig. 2, which reflects the stages of elimination and article selection about the PRISMA methodology. This selection process was carried out carefully and carefully to ensure that the articles used in this research were truly relevant and contributed to the understanding of DEG evaluation tools. This process is carried out by systematic research methodology standards to ensure the accuracy and validity of the data that will be used in subsequent analyses result. The results of processing the number of journals extracted can be seen in Table I.

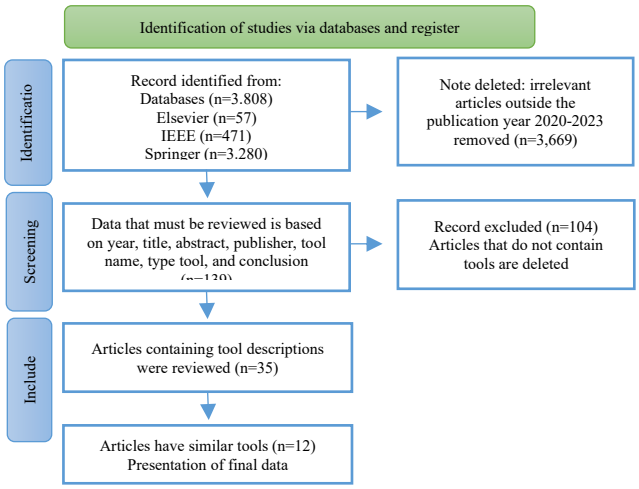



Fig. 2. Flow diagram of literature search and study selection [11]

## III. Result

The evaluation conducted in this study involved multiple levels that had been developed based on Kirkpatrick's four-level model, which has been the basis of evaluation for more than 40 years. This model presents a comprehensive framework for evaluating the effectiveness of educational interventions and partnerships at four levels: reaction, learning, behavior, and outcomes [11]. In this research, the evaluation was carried out by considering four things: model,

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TABLE I. CHARACTERISTICS OF TOOL NAME SIMILARITIES

| References | Year of Publication | Publisher | Tool name | Type of tools | Evaluation factors |
|---|---|---|---|---|---|
| [13], [14], [15], [7], [16] | 2021, 2022, 2023 | IEEE, Springer Nature, Springer | ADDIE | Design Models | Analysis, Design, Development, Implementation, Evaluation |
| [17] | 2023 | IEEE | DIJS | Design Models | Design, Evaluation, Pedagogical, Profile |
| [18] | 2023 | IEEE | ECS | Design Models | Component, Entity, System |
| [19], [20], [21] | 2022 | IEEE | EDUGXQ | Design Models | Attraction, Challenge, Concentration, Control, Engagement, Engrossment, Feedback, Flow, Goals, Immersion, Knowledge improvement, Learnability, Learning Content, Learning goals, Operability, Playability Player Context, Prior experience, Satisfaction, system, Understandability Usability, User environment |
| [19], [20], [22], [23], [8], [24], [11] | 2021, 2022 | Elsevier | Egame Flow | Scale Evaluation | Action, Autonomy, Challenge Clarity, Concentration, Consciousness, Control Experience, Fantasy, Feedback, Fun, Goals, Immersion, Knowledge improvement, Learning Mystery, Player Skills, Sensation, Social Interaction, Time |
| [25], [26], [19], [20], [21] | 2021, 2022, 2023 | Springer, IEEE | GUESS | Scale Evaluation | Audio and Visual Aesthetics, Creative Freedom, Engrossment, Enjoyment, Narratives, Personal, Playability, Gratification, Social connectivity, Usability |
| [27], [28], [29], [30] | 2021, 2022 | Elsevier, Springer | HEP | Design Models | Acceptability, Accessibility, Aesthetics, Assessment, Clarity, Consistency, Content, Control, Edit, Effectiveness, Efficiency, Engagement, Error, Esthetic, Feedback, Flexibility, Game Play, Game Story, Help, Human-centric syntax, Learnability, Levels, Mechanics, Memorability, Non-threatening, Operability, Pedagogical, Recognition, redundancy, Satisfaction, Strategy, Support, System, Usability, Viscosity, Visibility |
| [19], [21], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40] | 2020, 2012, 2023 | Elsevier, Springer, IEEE | MEEGA+ | Design Models | Accessibility, Aesthetics, Attention, Challenge, Confidence, Focused attention, Fun, Learnability, Learning goals, Learning outcome, Operability, Perceived learning, Player experience, Recovery, Relevance, Satisfaction, Short term learning, Social interaction, Usability, User error, User error protection |
| [41], [42] | 2021, 2023 | Elsevier, IEEE | SPIKES | Design Models | Emotions, Invitation, Knowledge, Perception Setting, Summary |
| [43] | 2023 | IEEE | UEQ | Scale Evaluation | Attractiveness, Dependability, Efficiency, Novelty, Perspicuity, Stimulation |
| [19], [8], [21], [29], [30], [44], [45] | 2022 | Elsevier, IEEE, Springer | TAM | Acceptance Model | 3D graphics, Attitude, Concept Map Usefulness, Enjoyment, Design, Intention to Use (ITU), Interface design, Operation Function, Perceived Ease of Use (PEU), Perceived Usefulness (PU), Satisfaction, Social influence |
| [19], [20], [21], [45] | 2022 | Elsevier, IEEE | UTAUT | Acceptance Model | Anxiety, Behavioral Intention to Use, Effort expectancy, Facilitating conditions, Hedonic motivation, Negative emotions, Performance expectancy, Personal innovativeness, Positive emotions, Value, Social influence |

scale, questionnaire, and technology acceptance. Titles and abstracts were screened independently based on eligibility criteria. Full texts were independently retrieved and screened by four authors based on eligibility criteria. To produce data that is more specific and easy to understand, data extraction was carried out by focusing on only a few variables, namely: reference, publication year, publisher, tool name, tool type, and evaluation factors.

A total of 35 produced 157 evaluation factors from all evaluation tools, 3 types of evaluation tools (Design Model, Scale Evaluation, Acceptance Model), and 12 names of DEG evaluation tools, namely ADDIE ([13]-[14]-[15]-[7]-[16]), DIJS [17], ECS [18], EDUGXQ ([19]-[20]-[21]), EgameFlow ([19]-[20]-[22]-[23]-[8]-[24]-[11]), GUESS ([25]-[26]-[19]-[20]-[21]), HEP ([27]-[28]-[29]-[30]), MEEGA+ ([19]-[21]-[31]-[32]-[33]-[34]-[35]-[36]-[37]-[38]-[39]-[40]), SPIKES ([41]-[42] ), UEQ ([43]), TAM ([19]-[8]-[21]-[29]-[30]-[44]-[45]), dan UTAUT ([19]-[20]-[21]-[45]).

The next process is giving a score based on the Analytical Hierarchy Process (AHP), where the scale used is 1-9, interpreted in TABLE II.

TABLE II. CHARACTERISTICS OF TOOL NAME SIMILARITIES

| Scale | Meaning |
|---|---|
| 1 | Comparing the two factors, both are equally important |
| 3 | Comparing the two factors, the former is slightly more important than the latter |
| 5 | Comparing the two factors, the former is more important than the latter |
| 7 | Comparing the two factors, the former is certainly more important than the latter |
| 9 | Comparing the two factors, the former is much more important than the latter |
| 2,4,6,8 | The judgment is between the two adjacent judgments |
| reciprocal | If the ratio of element it's importance to element j's importance is $a_{ij}$ , then the ratio of element j's importance to element i's importance is $a_{ji}=1/a_{ij}$ |

Relatives are entered into the formula to test whether the index weights are reasonable. Consistency index (CI) and consistency ratio (CR) are important values in AHP consistency testing. Calculation of the largest characteristic root λmax, where n is the number of indices, as shown in the following equation:

$$\lambda_{max} = \frac{1}{n}\sum_{i=1}^{n}\frac{(AW)_i}{W_i} \quad (1)$$

A is the scoring matrix, W=[w1,w2,…,wn]T,(AW)i is the i-th element after the matrix operation between A and W, and wi represents the index weights [46].

TABLE 3. LEVEL OF USE OF EVALUATION AND SCORING TOOLS

| No | Nama Alat | Jumlah Paper | Prosentase | Scoring |
|---|---|---|---|---|
| 1 | MEEGA+ | 12 | 23 % | 9 |
| 2 | EGameFlow | 7 | 13 % | 7 |
| 3 | TAM | 7 | 13 % | 5 |
| 4 | ADDIE | 5 | 10 % | 3 |
| 5 | GUESS | 5 | 10 % | 3 |
| 6 | HEP | 4 | 8 % | 3 |
| 7 | UTAUT | 4 | 8 | 3 |
| 8 | EDUGXQ | 3 | 6 | 1 |
| 9 | SPIKES | 2 | 4 | 1 |
| 10 | UEQ | 1 | 2 | 1 |
| 11 | DIJS | 1 | 2 | 1 |
| 12 | ECS | 1 | 2 | 1 |

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The level of evaluation and assessment tool used in journal extraction results can be presented in detail in numerical form. The table provides a structured overview of the more popular evaluation tools used in systematic evaluation studies of DEGs, namely MEEGA+, with 12 or 23% journal references. The level of use of evaluation and assessment tools is more clearly presented in Table III. which is arranged sequentially.

Based on the evaluation factor similarity data in this review, 28 similarity evaluation factors are shown in Fig. 3, while evaluation factors that have nothing in common with other evaluation tools are not displayed. From this, it can be seen that satisfaction and usability evaluation factors are more widely used in the evaluation, namely, respectively, 4 tools each. The characteristics of the evaluation factor equation can be seen in Table IV.

TABLE IV. CHARACTERISTICS OF EVALUATION FACTOR EQUATIONS

| No | Evaluation factors | Tool Name | Quantity | Priority Scale |
|---|---|---|---|---|
| 1 | Satisfaction | EDUGXQ, HEP, **MEEGA+**, TAM | 4 | - |
| 2 | Usability | EDUGXQ, HEP, GUESS, **MEEGA+** | 4 | - |
| 3 | Challenge | EDUGXQ, EgameFlow, **MEEGA+** | 3 | - |
| 4 | Learnability | EDUGXQ, HEP, **MEEGA+** | 3 | - |
| 5 | Operability | EDUGXQ, HEP, **MEEGA+** | 3 | - |
| 6 | Experience | EDUGXQ, EgameFlow, **MEEGA+** | 3 | - |
| 7 | Control | EDUGXQ, EgameFlow, HEP | **3** | 1 |
| 8 | Feedback | EDUGXQ, EgameFlow, HEP | **3** | 2 |
| 9 | Knowledge | EDUGXQ, EgameFlow, SPIKES | **3** | 3 |
| 10 | Design | ADDIE, DIJS, TAM | **3** | 4 |
| 11 | System | ECS, EDUGXQ, HEP | **3** | 5 |
| 12 | Accessibility | HEP, MEEGA+ | 2 | 6 |
| 13 | Aesthetics | HEP, MEEGA+ | 2 | 7 |
| 14 | Clarity | EgameFlow, HEP | 2 | 8 |
| 15 | Concentration | EDUGXQ, EgameFlow | 2 | 9 |
| 16 | Efficiency | HEP, UEQ | 2 | 10 |
| 17 | Engagement | EDUGXQ, HEP | 2 | 11 |
| 18 | Engrossment | EDUGXQ, GUESS | 2 | 12 |
| 19 | Enjoyment | GUESS, TAM | 2 | 13 |
| 20 | Evaluation | ADDIE, DIJS | 2 | 14 |
| 21 | Fun | EgameFlow, MEEGA+ | 2 | 15 |
| 22 | Goals | EDUGXQ, EgameFlow | 2 | 16 |
| 23 | Immersion | EDUGXQ, EgameFlow | 2 | 17 |
| 24 | Learning | EgameFlow, MEEGA+ | 2 | 18 |
| 25 | Learning goals | EDUGXQ, MEEGA+ | 2 | 19 |
| 26 | Pedagogical | DIJS, HEP | 2 | 20 |
| 27 | Perceived | MEEGA+, TAM | 2 | 21 |
| 28 | Playability | EDUGXQ, GUESS | 2 | 22 |

From Table 4, the selection of evaluation tools in this study is based on the most relevant and frequently used evaluation factors found in the initial analysis. MEEGA+ is one of the most frequently used evaluation tools because it has six main evaluation factors: Satisfaction, Usefulness, Challenge, Learnability, Operation, and Experience. These factors cover important aspects in evaluating digital educational games and are relevant in various studies.

However, MEEGA+ does not cover several other evaluation factors that are also considered important in the context of this research, namely Control, Feedback, Knowledge, and Design. Because these factors were not included in MEEGA+ but were identified as important by other evaluation tools, they were prioritized in the subsequent evaluation process. This ensures that all critical aspects in evaluating digital educational games are considered, resulting in a more comprehensive and holistic analysis.

Thus, the evaluation tools included are those with at least three evaluation factors in common with other evaluation tools and those that do not have certain critical evaluation factors such as Control, Feedback, Knowledge, and Design. This approach allows research to exploit the strengths of various evaluation tools while addressing their respective shortcomings, thereby ensuring that the evaluations conducted cover all relevant and significant aspects in the context of digital educational games.

To make it easier to understand, the evaluation factor equation can be visually presented in the form of a Search Gap, as can be seen in Fig. 3.

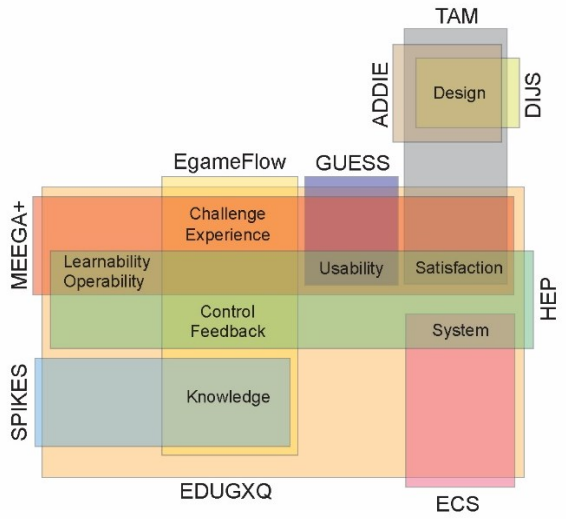



Fig. 3. Research gap

From Fig. 3 above, it can be seen that EDUGXQ's evaluation factors have the most similarities used by other evaluation tools; apart from that, it can be seen that no evaluation tool has evaluation factors that are used in all evaluation tools, meaning that there is the possibility of an evaluation tool being developed to be more complete and produce a more valid evaluation.

### *A. DEG design evaluation model*

- Tan et al. [47] evaluated using questionnaires and interviews to explore the learning conditions of underachieving students using a game concept platform, GamiClass, by referring to the ADDIE and Grasha Learning Styles instructional design models. The learning motivation of most students who fall into the underachieving category and actively participate in this game concept platform has increased significantly.
- Another relevant method is the Model for the Evaluation of Educational Games (MEEGA), which was specifically designed to assess DEG. This model uses a standard questionnaire that focuses on Kirkpatrick assessments, paying attention to student responses after interaction with the game [11]. MEEGA measures three qualitative dimensions of DEG, namely motivation, user experience, and learning outcomes, seen from students' perspectives in an educational context. MEEGA+ is an extension of the MEEGA model, which was developed through a review of the original version of the literature and systematic analysis [48].
- Desurvire et al. [29] developed a Heuristic Evaluation for Playability (HEP) based on a systematic review. This tool has been reviewed by several game experts and game designers. This tool categorizes evaluations into four dimensions: gameplay, game story, mechanics, and usability. Ultimately, alternative solutions are created by evaluators and game designers to solve game problems.
- Omari et al. [10] have developed a Fuzzy Multi-Criteria Decision Making (FMCDM) method to validate the

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selection of various options and give weight to the various criteria involved in the serious game evaluation process [48]. By applying the fuzzy TOPSIS and fuzzy ELECTRE methods to determine rankings on evaluation dimensions, they present an alternative evaluation model for serious games, which is different from the fuzzy AHP method approach.

### B. DEG Evaluation Scale

- Weng et al. [49], colleagues conducted research focused on ARTé Mecenas, a tool designed to support students' art history learning. In this research, the GameFlow Model is used. The research results show that ARTé: Mecenas can provide an entertaining gaming experience for players. The advantage of adopting the EGameFlow scale lies in considering the learning aspects contained in the game.

- Another scale provided by Silveira et al. [50], aims to measure user satisfaction with video games using nine scales. This scale is known as the Educational Game User Experience Satisfaction (E-GUESS) user experience scale. E-GUESS has been proven to have content validity, internal consistency, convergent validity, and differentiation.

### C. Other DEG Evaluation Approaches (questionnaires)

- Tori et al. [51], in their research, explained that the Pre/Post-test questionnaire is the most commonly used among various user performance evaluation tools. Pre/Post-test questionnaires are often designed to ascertain whether learning occurred or achieved positive results with the proposed system. Most scientific articles list the instruments used to validate their approach, and pre/post-test questionnaires are becoming a more common mechanism researchers apply.

### D. Technology Acceptance Model

- Rouidi et al. [44] explained the technology acceptance model TAM (Technology Acceptance Model) and UTAUT, which aims to understand better why users accept or reject certain technologies and how user acceptance can be improved through technology design and can measure post-acceptance and with additional factors [45].

- Razami and Ibrahim [8] presented a systematic literature review regarding predictors of DEG receipt in their study. In this research, it is stated that several studies have used the technology acceptance model (TAM) as a tool to evaluate students' intentions to utilize DEG as a learning tool. Construct factors influencing intention include perceived ease of use, perceived usefulness, level of enjoyment, attitude, satisfaction, and social influence. In this paper, emphasis is also placed on the models and constructions used to predict DEG acceptance [19].

- Pilapaxi-Cunalata and Llerena-Izquierdo [52], in their research, concluded that the UTAUT Model is an effective tool for assessing student acceptance levels after applying various technological tools in the context of the teaching and learning process. This model links and integrates key elements applied in various contexts using digital technologies, emphasizing educational environments.

## IV. Discussion

The study's analysis provided 35 articles containing educational game evaluation tools. Based on this data, classification was then carried out based on the similarity of tool names. From the classification of tool name similarities, 12 names of tools were obtained that were used in evaluating educational games, namely ADDIE, DIJS, EDUGXQ, EgameFlow, GUESS, HEP, MEEGA+, SPIKES, UEQ, TAM, and UTAUT.

The results obtained from the tool equation classification process showed that MEEGA+ was the evaluation tool that was more widely used [11] in the journals obtained, namely 12 references or 23%, followed by EgameFlow and others.

The evaluation factors were then searched for similarities with the table, and it was found that satisfaction and usability were more widely used in the evaluation factors, namely 4 tools each. Based on the order of quantity of tool names used, MEEGA+ is spread across the seven largest orders, meaning that MEEGA+ evaluation factors are more similar to evaluation factors in other evaluation tools, so it is often referred to as a reliable evaluation tool [31]. Still, with the development of GED and manufacturing technology, As games become easier and faster, it is necessary to develop continuous evaluation games so that they can produce increasingly valid evaluation models [11].

The next step is to carry out a weighting process using AHP to determine the priority order of which evaluation factors will be used to develop MEEGA+, which is considered more popular in this process. From the calculation results it was found that control and feedback got the same high score, namely three. According to Mayuri Borah, control is one of the important aspects that can increase student motivation in learning namely students feel they can control by seeing a direct relationship between their actions and results, besides that the control aspect can also produce a positive response and make the mood happy. playing digital games [53] so it will be developed to be used as an evaluation factor combined with MEEGA+. This systematic review shows the need to develop a more complete evaluation model for educational games in different periods [11] so that it can be used to measure the quality of educational games more validly and can be used as a basis that can be used as a basis for deciding on the best use of DEG.

Suggestions for future research include the development of new evaluation tools or modifying existing tools based on the gaps identified in this research, which could provide significant benefits in several aspects. First, developing new evaluation tools that incorporate evaluation factors that are not yet comprehensively covered in existing tools, such as control and feedback, will increase the validity and reliability of educational game evaluations. Second, modifying existing tools, such as MEEGA+, to include new evaluation factors identified as important would make them more holistic and adaptive to changing technology and evolving learning methods.

In addition, more comprehensive evaluation tools will help educational researchers and practitioners identify the weaknesses and strengths of various educational games more precisely, thereby informing the development of more effective and efficient games. Using methods such as AHP for weighting evaluation factors also provides a systematic

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approach to determining priorities, which can improve the quality of decisions taken based on evaluation results.

Overall, these suggestions are expected to encourage further, more in-depth, and innovative research in the field of educational game evaluation, which will ultimately contribute to improving the quality of learning through digital game media.

## V. CONCLUSION

Based on the journal processing results obtained, it can be concluded that the distribution of journals by publisher shows Springer's dominance at 86%, followed by IEEE at 12%, and Elsevier at 2%. There are 12 evaluation tools obtained from the processed journals, namely MEEGA+, EgameFlow, TAM, ADDIE, GUESS, HEP, TAUT, EDUGXQ, SPIKES, UEQ, DIJS, and ECS. Despite variations in evaluation tools, MEEGA+ proved to be the more popular evaluation model used across the 12 journals. Journal processing also revealed 9 similarities in evaluation tools and 28 similarities in evaluation factors in existing evaluation tools. However, it should be noted that no evaluation tool covers all evaluation factors. This conclusion emphasizes the complexity and diversity of DEG evaluation tools, thereby creating challenges for developing comprehensive and inclusive evaluation tools in future research.

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